

# Task 2: Understand Android Studio and App Structure

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## Objectives

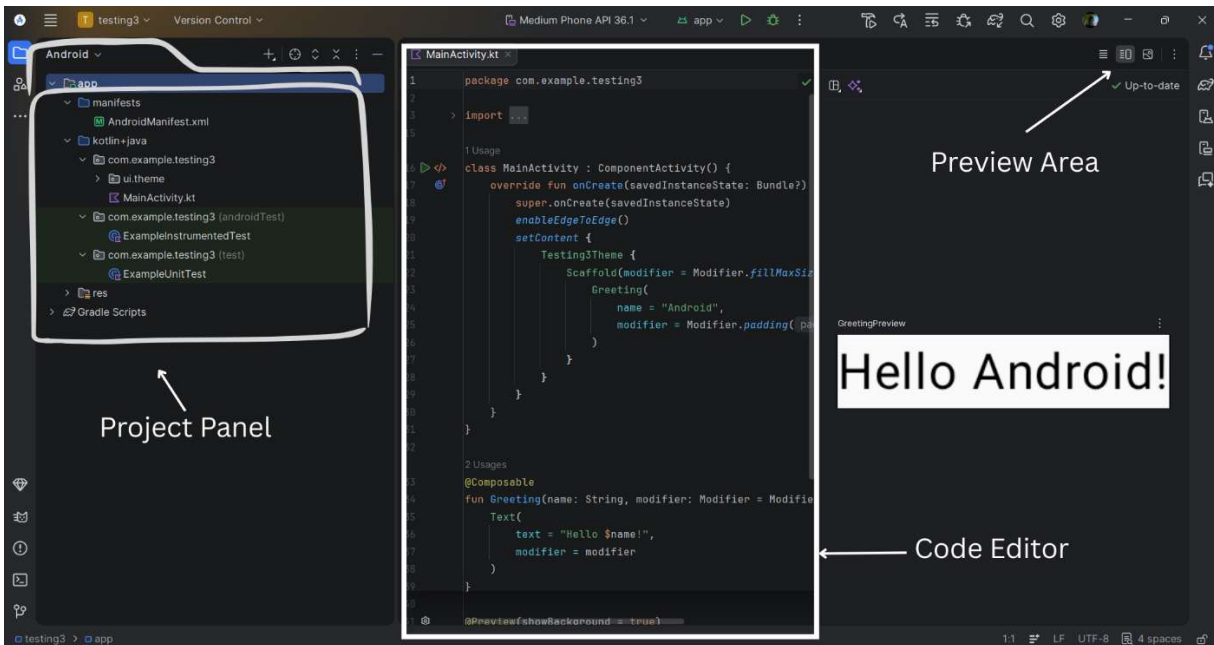
- To understand the Android Studio development environment and its interface.
- To identify and explore key components such as the Project Panel, Code Editor, and Preview Area.
- To recognize important project files like MainActivity.kt, AndroidManifest.xml, strings.xml, and build.gradle (Module).
- To understand how different files and resources work together to build and run an Android application.
- To develop basic documentation and diagramming skills for explaining project structure clearly.

## Introduction

Android Studio is the official Integrated Development Environment (IDE) for Android app development. It provides tools for writing code, managing resources, previewing user interfaces, and configuring application settings. Every Android project follows a structured organization that includes source files, resource folders, manifest configurations, and build scripts.

Understanding how files such as MainActivity.kt, AndroidManifest.xml, res/values/strings.xml, and build.gradle (Module) interact is essential before writing code. A clear grasp of the project structure improves efficiency, reduces errors, and helps developers manage applications systematically. This task focuses on exploring the environment, documenting key components, and analyzing how they connect to produce a working Android app.

## 1. Navigate Android Studio:



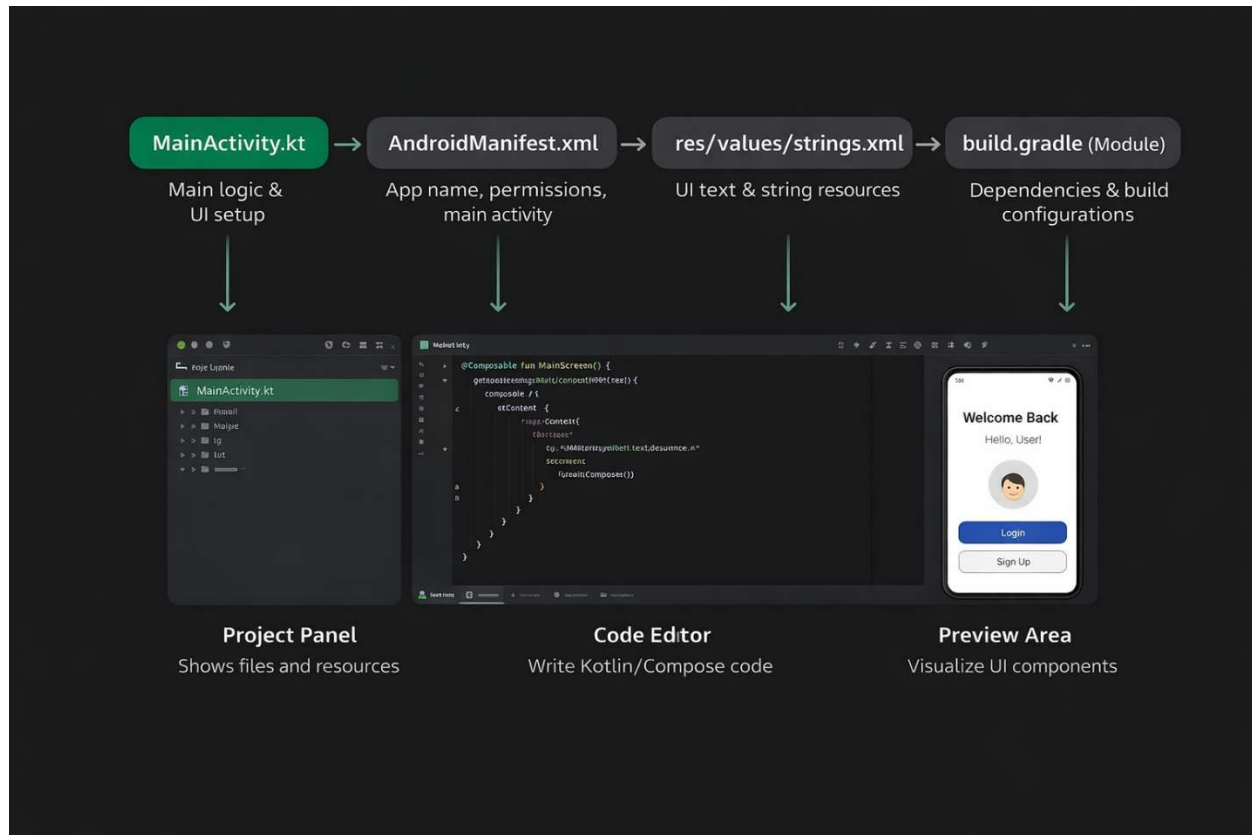
Canva link:

[https://www.canva.com/design/DAHBnC\\_0V5s/dtEZaEUPMHckidpCoyWCdA/view?utm\\_content=DAHBnC\\_0V5s&utm\\_campaign=designshare&utm\\_medium=link2&utm\\_source=uniqueLinks&utm\\_id=h27216d222e](https://www.canva.com/design/DAHBnC_0V5s/dtEZaEUPMHckidpCoyWCdA/view?utm_content=DAHBnC_0V5s&utm_campaign=designshare&utm_medium=link2&utm_source=uniqueLinks&utm_id=h27216d222e)

## 2. Explore Project Structure:

| File Name                  | Location in Project                                 | Purpose in the Project                                                                                                      |
|----------------------------|-----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| MainActivity.kt            | app > java ><br><package_name> ><br>MainActivity.kt | Contains the main application logic and UI setup using Jetpack Compose. Acts as the entry point of the app.                 |
| AndroidManifest.xml        | app > manifests ><br>AndroidManifest.xml            | Defines the app name, permissions, main launcher activity, and overall application configuration.                           |
| res/values/strings.xml     | app > res > values ><br>strings.xml                 | Stores all string resources used in the app such as app name, labels, and messages. Helps in localization and organization. |
| build.gradle (Module: app) | app > build.gradle or app ><br>build.gradle.kts     | Manages dependencies, SDK versions, plugins, and build configurations for the application module.                           |

### 3. File Mapping Diagram:



### 4. Mini Reflection:

**“ How Android Studio organizes a project and why understanding this structure is important before coding. ”**

1. Android Studio organizes projects into structured folders like java/kotlin, manifests, res, and Gradle scripts, each serving a specific purpose.
2. Files such as MainActivity.kt, AndroidManifest.xml, strings.xml, and build.gradle work together to manage logic, configuration, resources, and dependencies.
3. The Project Panel shows all files and resources, the Code Editor is used for writing code, and the Preview Area displays the UI output.
4. Understanding this structure helps in locating files quickly and managing the app efficiently.
5. Knowing the project organization before coding reduces errors and makes development smoother and more systematic.